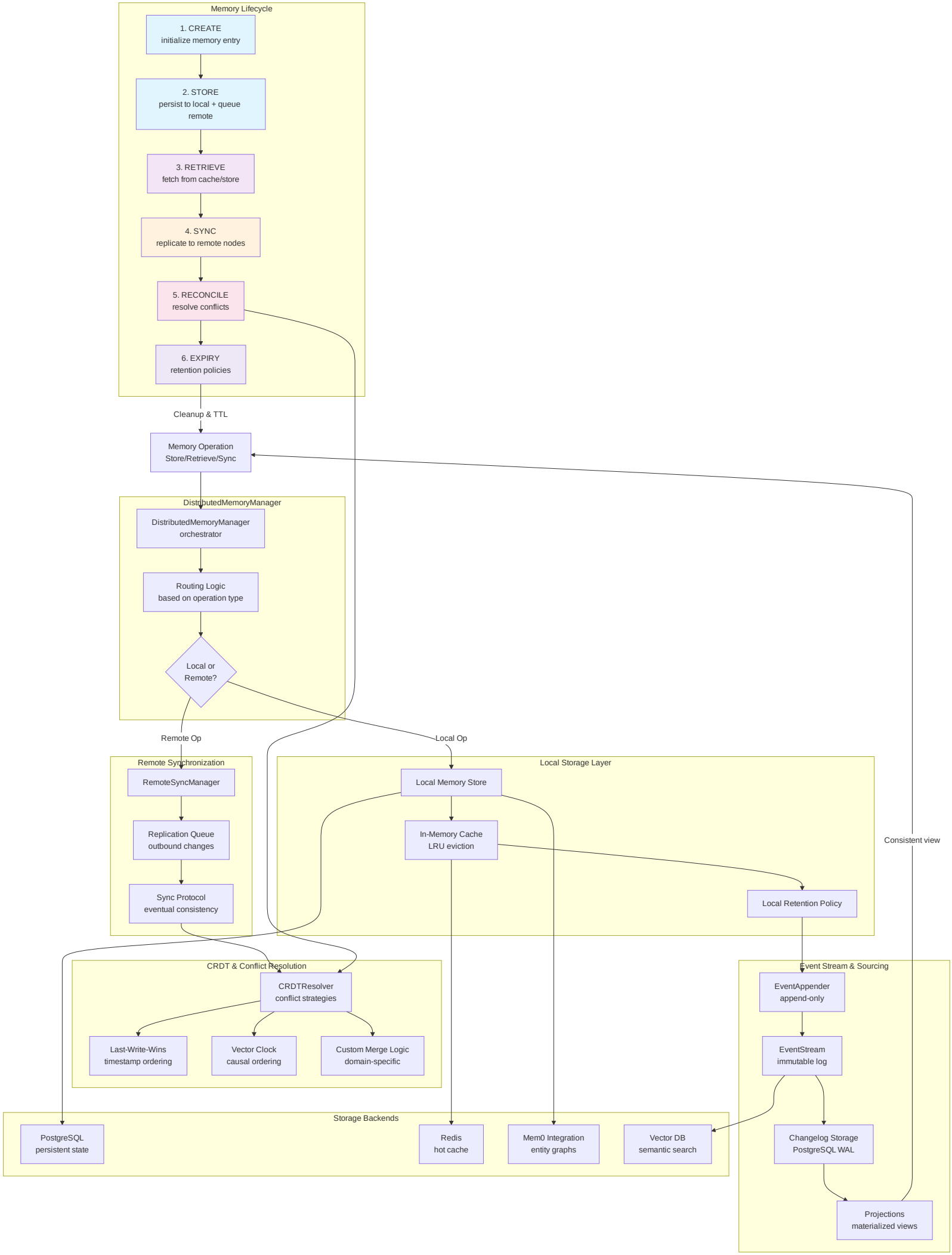
Memory Lifecycle
1. CREATE
initialize memory entry
2. STORE
persist to local + queue
remote
3. RETRIEVE
fetch from cache/store
4. SYNC
replicate to remote nodes
5. RECONCILE
resolve conflicts
6. EXPIRY
retention policies
Cleanup & TTL
Memory Operation
Store/Retrieve/Sync
DistributedMemoryManager
DistributedMemoryManager
orchestrator
Routing Logic
based on operation type
Local or
Remote?
Remote Op
Local Op
Remote Synchronization
RemoteSyncManager
Replication Queue
outbound changes
Sync Protocol
eventual consistency
Local Storage Layer
Local Memory Store
In-Memory Cache
LRU eviction
Local Retention Policy
Consistent view
CRDT & Conflict Resolution
CRDTResolver
conflict strategies
Last-Write-Wins
timestamp ordering
Vector Clock
causal ordering
Custom Merge Logic
domain-specific
Event Stream & Sourcing
EventAppender
append-only
EventStream
immutable log
Changelog Storage
PostgreSQL WAL
Projections
materialized views
Storage Backends
PostgreSQL
persistent state
Redis
hot cache
Mem0 Integration
entity graphs
Vector DB
semantic search